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STATISTICS

Fourth Edition

David Freeman, Robert Pisani, and Roger Purves

Statistics

Fourth Edition

DAVID FREEDMAN
ROBERT PISANI
ROGER PURVES

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Fourth Edition

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To Jerzy Neyman (1894–1981)

Born in Russia, Neyman worked in Poland and England before coming to the United States in 1938. He was one of the great statisticians of our time.

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Preface

What song the Sirens sang, or what name Achilles assumed when he hid among women, though puzzling questions, are not beyond all conjecture.

—SIR THOMAS BROWNE (ENGLAND, 1605–1682)

TO THE READER

We are going to tell you about some interesting problems which have been studied with the help of statistical methods, and show you how to use these methods yourself. We will try to explain why the methods work, and what to watch out for when others use them. Mathematical notation only seems to confuse things for many people, so this book relies on words, charts, and tables; there are hardly any x 's or y 's. As a matter of fact, even when professional mathematicians read technical books, their eyes tend to skip over the equations. What they really want is a sympathetic friend who will explain the ideas and draw the pictures behind the equations. We will try to be that friend, for those who read our book.

WHAT IS STATISTICS?

Statistics is the art of making numerical conjectures about puzzling questions.

- What are the effects of new medical treatments?
- What causes the resemblance between parents and children, and how strong is that force?
- Why does the casino make a profit at roulette?
- Who is going to win the next election? by how much?
- How many people are employed? unemployed?

These are difficult issues, and statistical methods help a lot if you want to think about them. The methods were developed over several hundred years by people who were looking for answers to their questions. Some of these people will be introduced later.

AN OUTLINE

Part I is about designing experiments. With a good design, reliable conclusions can be drawn from the data. Some badly-designed studies are discussed too—so you can see the pitfalls, and learn what questions to ask when reading about a study. Study design is perhaps our most important topic; that is why we start there. The ideas look simple, but appearances may be deceptive: part I has a lot of depth.

Studies typically produce so many numbers that summaries are needed. Descriptive statistics—the art of summarizing data—is introduced in part II. Histograms, the average, the standard deviation, and the normal curve are all considered. The discussion continues in part III, where the focus is on analyzing relationships, for instance, the dependence of income on education. Here, correlation and regression are the main topics.

Much statistical reasoning depends on the theory of probability, discussed in part IV; the connection is through chance models, which are developed in part V. Coins, dice, and roulette wheels are the main examples in parts IV and V. The expected value and standard error are introduced; probability histograms are developed, and convergence to the normal curve is discussed.

Statistical inference—making valid generalizations from samples—is the topic of parts VI–VIII. Part VI is about estimation. For instance, how does the Gallup Poll predict the vote? Why are some methods for drawing samples better than others? Part VII uses chance models to analyze measurement error, and to develop genetic theory. Part VIII introduces tests of significance, to judge whether samples are consistent with hypotheses about the population. As parts VI–VIII show, statistical inferences depend on chance models. If the model is wrong, the resulting inference may be quite shaky.

Nowadays, inference is the branch of statistics most interesting to professionals. However, non-statisticians often find descriptive statistics a more useful branch, and the one that is easier to understand. That is why we take up descriptive statistics before inference. The bare bones of our subject are presented in chapters 1 to 6, 13, 16 to 21, 23, and 26. After that, the reader can browse anywhere. The next chapters to read might be 8, 10, 27, and 29.

EXERCISES

The sections in each chapter usually have a set of exercises, with answers at the back of the book. If you work these exercises as they come along and check the answers, you will get practice in your new skills—and find out the extent to which you have mastered them. Every chapter (except 1 and 7) ends with a set of review exercises. The book does not give answers for those exercises. Chapters 6, 15, 23, and 29 also have “special review exercises,” covering all previous material. Such exercises must be answered without the clues provided by context.

When working exercises, you might be tempted to flip backward through the pages until the relevant formula materializes. However, reading the book backward will prove very frustrating. Review exercises demand much more than formulas. They call for rough guesses and qualitative judgments. In other words, they require a good intuitive understanding of what is going on. The way to develop that understanding is to read the book forward.

Why does the book include so many exercises that cannot be solved by plugging into a formula? The reason is that few real-life statistical problems can be solved that way. Blindly plugging into statistical formulas has caused a lot of confusion. So this book teaches a different approach: thinking.

GRAPHICS

As in previous editions, extensive use is made of computer graphics to display the data. Working drawings, however, are done freehand; the reader is encouraged to make similar sketches, rather than being intimidated by too much precision. The book still features cartoons by Dana Fradon of *The New Yorker*.

What's New in the Fourth Edition?

Of the making of books, there is no end.
—*Ecclesiastes*

The principal change is to the data. Statistics, like people, show wear and tear from aging. Fortunately or unfortunately, data are easier to rejuvenate. We started the first edition in 1971, and completed the fourth in 2006. These past 35 years were years of rapid change, as commentators have doubtless observed since prehistoric times.

There was explosive growth in computer use. Other technical developments include email (+), the world wide web (+), Windows ($\pm$), cell phones ($\pm$), and call centers with voice-activated menus (–). SAT scores bottomed out around 1990, and have since been slowly going up (chapter 5). Educational levels have been steadily increasing (chapter 4), but reading skills may—or may not—be in decline (chapter 27).

The population of the United States increased from 200 million to 300 million (chapter 24). There was corresponding growth in higher education. Over the period 1976 to 1999, the number of colleges and universities increased from about 3,000 to 4,000 (chapter 23). Student enrollments increased by about 40%, while the professoriate grew by 60%. The number of male faculty increased from 450,000 to 600,000; for women, the increase was 175,000 to 425,000. Student enrollments shifted from 53% male to 43% male.

There were remarkable changes in student attitudes (chapters 27, 29). In 1970, 60% of first-year students thought that capital punishment should be abolished; by 2000, only 30% favored abolition. In 1970, 36% of them thought that “being very well off financially” was “very important or essential”; by 2000, the figure was 73%.

The American public gained a fraction of an inch in height, and 20 pounds in weight (chapter 4). Despite the huge increase in obesity, there were steady gains in life expectancy—about 7 years over the 35-year period. Gain in life expectancy is a process (“the demographic transition”) that started in Europe around 1800. The trend toward longer lives has major societal implications, as well as ripple effects on our exercises.

Family incomes went up by a factor of four, although much of the change represents a loss of purchasing power in the dollar (chapter 3). Crime rates peaked somewhere around 1990, and have fallen precipitously since (chapters 2, 29). Jury awards in civil cases once seemed out of control, but have declined since the 1990s

along with crime rates. (See chapter 29; is this correlation or causation?) Our last topic is a perennial favorite: the weather. We have no significant changes to report (chapters 9, 24).*

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*Most of the data cited here come from the *Statistical Abstract of the United States*, various editions. See chapter notes for details. On trends in life expectancy, see Dudley Kirk, “Demographic transition theory,” *Population Studies* vol. 50 (1996) pp. 361–87.

PART I

Design of Experiments

